Phishing Detection System

An Application to prevent citizens from phishing attacks…

Phishing

- Phishing is the most commonly used social engineering and cyber attack.

- Through such attacks, the phisher targets naïve online users by tricking them into revealing confidential information, with the purpose of using it fraudulently.

- There are many types of phishing to lure people into phishing attacks such as deceptive phishing, spear phishing, whalling and smishing etc…

MISSION STATEMENT

Our mission is to give protect the users from the malicious phishing attacks targeted towards them to collect their confidential information or other monetary thefts.

Our Application

The proposed system doesn’t need any invoking or command, it is automatically called by default when the user is subjected to click any links.

The main advantage of this app is that is doesn’t works for a single platform but monitors the overall mobile applications and blocks the user from entering into malicious phishing websites forwarded to them in any platform. Eh.. WhatsApp, Facebook messagener, Instagram etc…

FEATURE SELECTION

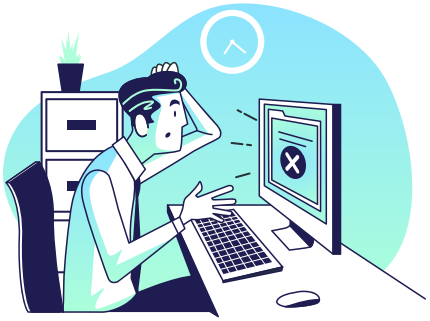
The following category of features are selected:

- Address Bar based Features
- Domain based Features
- HTML & Javascript based Feature
- Address Bar based Features considered are:
- Domian of URL • Redirection ‘//’ in URL
- IP Address in URL • ‘http/https’ in Domain name
- ‘@’ Symbol in URL • Using URL Shortening Service
- Length of URL • Prefix or Suffix “-” in Domain
- Depth of URL

FEATURE SELECTION

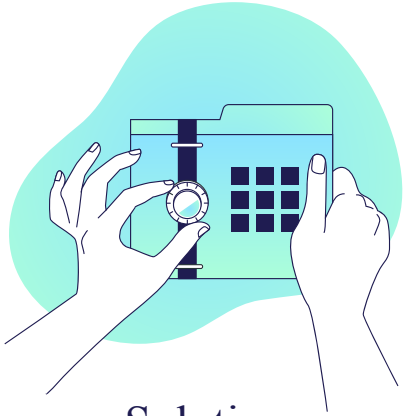
- Domain based Features considered are:
- HTML and JavaScript based Features considered are:
- DNS Record • Age of Domain
- Website Traffic • End Period of Domain
- Iframe Redirection • Disabling Right Click
- Status Bar Customization • Website Forwarding
- All together 17 features are extracted from the dataset.

PROBLEM VS. SOLUTION



Problem

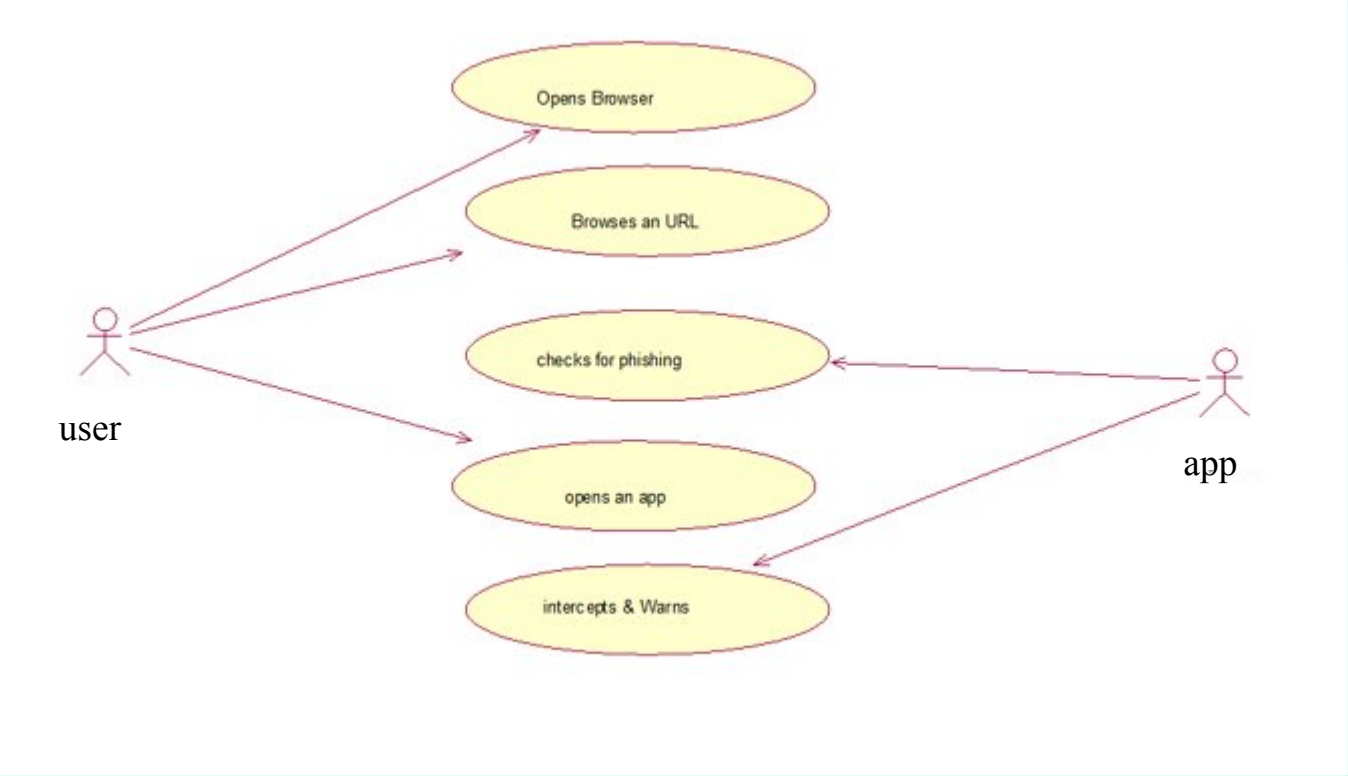
Phishing attacks that scams or deceive people into revealing their confidential information such as bank account, passwords etc…



Solution

Detection of phishing links pre-handedly before the user opens it.

USE CASE DIAGRAM



EXISTING SYSTEM

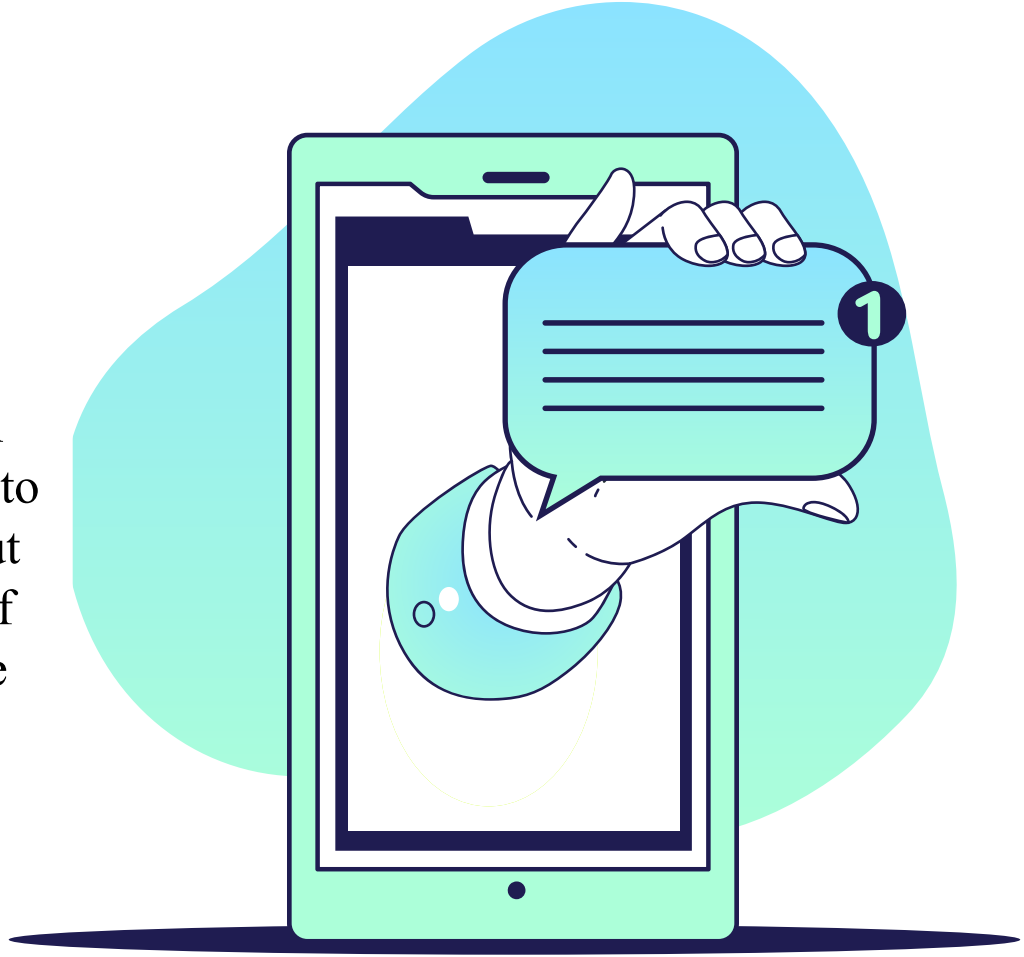
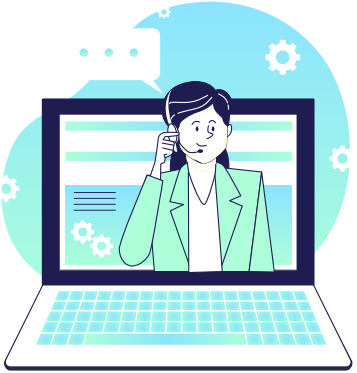
- The phishing detection system is confined to Google Chrome Browser. It uses Google's Safe Browsing Database to warn user before entering a possible phishing site.
- Although, it protects user from entering into phishing sites on Chrome Browser, it fails to do so on other browsers.
- It cannot detect newly created phishing sites (Someone has to report it)
- Doesn't work in other mobile browsers
- Doesn't scan all the links opened by the device(Mobile)
- Cannot intercept all network traffic like GET requests

PROPOSED SYSTEM

- To overcome the drawbacks of existing system, we propose a THE PHISHING application
- It is based on Machine learning classification model and can detect phishing attacks in mobile in realtime.
- Detects phishing attacks even on mobile devices
- It is not limited to one browser; can work on all browsers
- Not confined to browsers, scans links in other apps too.
- Detects all the network traffic and intercepts phishing sites.

Conclusion

By using RPADML system, we solve the problem of detecting phishing sites on mobile devices in real-time. Now, the users are able to identify phishing sites/links without interacting. RPADML system itself shows floating warning sign before entering such websites.



THANK YOU